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Influence of soil variability on soil-structure interaction

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ABSTRACT: This paper proposes a sub-structuring method to account for spatial variability of the soil parameters in the vicinity of the foundation in soil-structure interaction analysis. Spatial variability is modeled as a perturbation added to the soil modulii over a bounded part of the soil medium. The first step consists in the calculation of the Green function of the overall reference domain constituted by the structure and the soil without its perturbed part, and its impedance function when it is seen from the perturbed domain. The second step consists in solving the equilibrium equation of the last one. Expressions for the displacement fields in the soil and the structure, and the modified impedance functions of the foundation when spatial variability of the soil is accounted for are derived. The case of a random perturbation is finally addressed by introducing the Karhunen-Loeve expansion of the stiffness operator of the perturbed domain.

1 INTRODUCTION

Site effect and soil-structure interaction have received a broad attention in the recent years due to the significant role they play on the seismic response of massive structures such as nuclear power plants or arch dams. Site effect analysis are intended to provide the engineer with an improved model of the seismic free field compatible with the local site topographical irregularities and material heterogeneity in the vicinity of the contemplated structure (Geli et al. 1988). Soil-structure interaction computations are performed to take into account the diffraction and radiation of waves against and away from the base of the structure. Such studies provide the engineer with the seismic input motion applied to the structure foundation, which corresponds to the motion of the massless foundation without any superstructure, and the soil impedance functions (Wolf 1985, Clouteau 1990).

The development of the Boundary Element Method (BEM) implemented with a subdomain approach has made possible the analysis of both coupled concepts in a unified way and at reasonable computational costs even in the three-dimensional cases (see Aubry & Clouteau 1992 and references therein). Boundary integral formulations are particularly advantageous when applied to unbounded media, however their relevancy for heterogeneous domains is limited by the fact that only horizontally stratified materials can be considered; in such cases, the Green tensors of the medium are only known numerically (Clouteau 1990). The commonly stated assumption of lateral homogeneity of the soil underneath the foundation is not valid any more for large spread foundations whose impedance functions, as well as the seismic input motion, are expected to be significantly modified by the soil heterogeneities. Here we focus on the spatial variability of the soil modulii in the vicinity of the structure and the influence it may have on the structural response when a seismic incident field consisting in the superposition of plane waves impinges it. The influence of spatial variability of the free field has been addressed in an accompanying paper (Clouteau et al. 1998).

For complex shapes of the heterogeneity, boundary elements methods are classically coupled with finite elements methods in order to combine their respective advantages: flexibility of the discretization with FEM and allowance of wave radiation toward infinity in unbounded media with BEM. Various formulations have been proposed in the literature (refer for instance to the bibliography by Mackerle 1997). In this paper, a rather new approach is proposed which accounts for the heterogeneous part of the soil as a volume interface associated with the perturbed modulii of the soil medium. Recently, Muijres et al. (1998) have proposed a similar formulation for an acoustic medium with heterogeneities (elastic inclusions or cracks) having a small scale with respect to the wavelength. Comparisons with a Neumann expansion approach or self-consistent schemes for composite materials showed the advantages of such formulation. Here it is extended to the elastic case for a possibly large bounded heterogeneous soil part. The method couples the boundary element method with the finite element method for the perturbed domain. It renders possible the analysis of soil-structure interaction phenomena with deterministic or random heterogeneous inclusions underneath the foundation. We derive the expressions of the displacement in the structure or the soil, as well as the modified impedance functions.

2 NOTATIONS AND HYPOTHESIS

The elastic structure under investigation is identified by an open bounded set Ω_b. It has deterministic density ρ_b and Lamé's modulii λ_b and μ_b. The infinite visco-elastic soil domain is identified by two unbounded open sets, namely the reference model domain Ω_{s0} and the actual domain Ω_s (Fig. 1). They both have identical geometries, but different density ρ_s and Lamé's modulii λ_s and μ_s defined by:

$$\begin{aligned} &\gamma_s(\mathbf{x}) = \gamma_{s0}(\mathbf{x}) \qquad \forall \mathbf{x} \in \Omega_{s0} \\ &\gamma_s(\mathbf{x}) = \gamma_{s0}(\mathbf{x}) + \gamma_{s1}(\mathbf{x}) \qquad \forall \mathbf{x} \in \Omega_{s1} \\ &\gamma_s = \rho_s, \lambda_s \text{ or } \mu_s \end{aligned} \tag{1}$$

Ω_{s0} is the unbounded soil domain that would remain in the absence of the heterogeneity, with known parameters $(\rho_{s0},\lambda_{s0},\mu_{s0})$. Ω_{s1} is the bounded perturbed soil domain actually occupied by the heterogeneity in the vicinity of the structure foundation, with known (deterministic) or unknown (random) fluctuations $(\rho_{s1},\lambda_{s1},\mu_{s1})$. In the above it is supposed that $\gamma_{s1} = 0$ on $\partial\Omega_{s1}$. The actual and reference overall domains for the coupled system constituted by the structure and the soil are denoted $\Omega = \Omega_b \cup \Omega_s$ and $\Omega_0 = \Omega_b \cup \Omega_{s0}$ respectively. Γ_{ab} is the traction-free surface of the structure, Γ_{as} is the traction-free surface of the soil and Γ is the interface between the soil and structure domains. Finally, the radiation surface of the soil $\Gamma_{s\infty}$ is introduced for convenience.

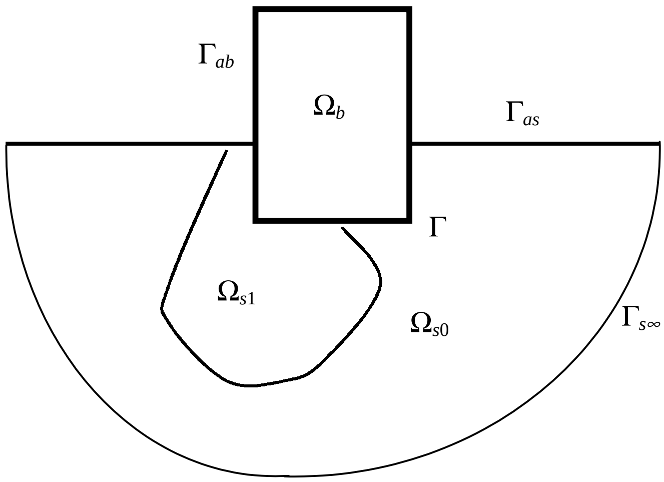

Figure 1. Geometry and notations for the problem.

For generality purposes, a seismic field impinging the structure is considered in this analysis. The incident free field in the reference domain, i.e. the seismic field which would propagate in the soil domain in the absence of its heterogeneous part Ω_{s1}, is denoted $\mathbf{u}^0_i(\mathbf{x},t)$ in the time domain. The overall problem being supposed linear, it is convenient to rather work in the frequency domain. Thus let $\mathbf{u}(\mathbf{x},\omega)$ be the displacement field on Ω induced by this incident field $\mathbf{u}^0_i(\mathbf{x},\omega)$ in the frequency domain, then it satisfies the homogeneous Navier equation in Ω:

$$\mathbf{Div}\boldsymbol{\sigma}(\mathbf{u}) = -\rho\omega^2\mathbf{u} \tag{2}$$

with traction-free boundary condition on $\Gamma_{as} \cup \Gamma_{ab}$ oriented by their outward normal unit vector $\mathbf{n}$:

$$\boldsymbol{\sigma}(\mathbf{u}) : \mathbf{n} = \mathbf{t_n}(\mathbf{u}) = \mathbf{0} \tag{3}$$

$\boldsymbol{\varepsilon}(\mathbf{u})$ and $\boldsymbol{\sigma}(\mathbf{u})$ are the strain and stress tensors and $\mathbf{t_n}(\mathbf{u})$ is the surface traction. The incident field consists in plane waves satisfying the homogeneous Navier equation in Ω_{s0} as well as the traction-free boundary condition on Γ_{as}. To ensure the uniqueness of the solution in the frequency domain, radiation conditions of the Sommerfeld's type on $\Gamma_{s\infty}$ have to be stated for the diffracted fields in the soil domain. These fields will be introduced later on, however it has to be noted here that the incident field in the reference domain as defined above does not satisfy these conditions.

In the following subscripts $(.)^0$ and $(.)^1$ will stand for any quantity calculated with the modulii of the reference soil domain $(\rho_{s0},\lambda_{s0},\mu_{s0})$ and the perturbed soil domain $(\rho_{s1},\lambda_{s1},\mu_{s1})$ respectively. No subscript corresponds to the actual domain. Finally the notations below hold for two vector fields $\mathbf{u}$ and $\mathbf{v}$ on a given domain Ω_a with the Euclidean scalar product $\mathbf{u}.\mathbf{v}$ and its norm $|\mathbf{u}|$:

$$(\mathbf{u},\mathbf{v})_{\Omega_a} = \int_{\Omega_a} \mathbf{u}(\mathbf{x}).\mathbf{v}(\mathbf{x})dV(\mathbf{x})$$

or their trace on a given surface Γ_a:

$$\langle \mathbf{u},\mathbf{v}\rangle_{\Gamma_a} = \int_{\Gamma_a} \mathbf{u}(\mathbf{x}).\mathbf{v}(\mathbf{x})dS(\mathbf{x})$$

For two tensors $\boldsymbol{\varepsilon}$ and $\boldsymbol{\sigma}$ we let:

$$(\boldsymbol{\varepsilon},\boldsymbol{\sigma})_{\Omega_a} = \sum_{i,j}\int_{\Omega_a} \varepsilon_{ij}(\mathbf{x}).\sigma_{ij}(\mathbf{x})dV(\mathbf{x})$$

The associated norms are noted $|\mathbf{u}|_\Sigma$ with:

$$\|\mathbf{u}\|_\Sigma = \langle \mathbf{u},\bar{\mathbf{u}}\rangle_\Sigma$$

Σ being a surface or a volume and $\bar{\mathbf{u}}$ denoting the complex conjugate of $\mathbf{u}$.

3 GREEN FUNCTION OF THE OVERALL REFERENCE MEDIUM

A model, which is able to account for the soil-structure interaction and homogeneous soil-perturbed soil interaction, has to be build. In this respect, a sub-structuring approach is proposed here. The present method is inherited from classical dynamic sub-structuring methods used for soil-structure or fluid-structure interaction problems. They are extended here to the case of a so-called "volume interface" in between the homogeneous and perturbed parts of the soil domain: it is geometrically defined by the volume occupied by the heterogeneity of the soil and mechanically it has modulii equal to $(\rho_{s1},\lambda_{s1},\mu_{s1})$. The relevancy of such a definition is apparent if we refer to the classical perturbation methods where the stiffness of the heterogeneous part of the propagation medium is represented as an additional force exerted on the unperturbed reference medium.

The unperturbed reference medium is chosen as the overall reference coupled domain Ω_0. We introduce its Green function noted $\mathbf{u}^0_g(\mathbf{x},\omega;\boldsymbol{\xi},\mathbf{e})$ which represents the displacement at the circular frequency ω and at the point $\mathbf{x}$ when a unit load is applied at the location $\boldsymbol{\xi}$ along the direction $\mathbf{e}$ in the soil or in the structure. Let $(\mathbf{e}_1,\mathbf{e}_2,\mathbf{e}_3)$ be an orthonormal basis of 3^3. Then for any given force density $\mathbf{q}$ applied to a region Ω_q in the reference soil medium, the displacement in the overall reference domain is given at any point **x** by the integral representation:

$$\mathbf{u}^0(\mathbf{x},\omega) = \sum_{k=1}^{3} \int_{\Omega_q} \mathbf{u}_g^0(\mathbf{x},\omega;\boldsymbol{\xi},\mathbf{e}_k)(\mathbf{q}(\boldsymbol{\xi}).\mathbf{e}_k)dV(\boldsymbol{\xi}) \tag{4}$$

We first present the method developed to compute this Green function by a sub-structuring technique: two elastodynamic sub-problems are dealt with, and afterwards coupled writing the equilibrium of the soil-structure interface in the weak sense. Then it is shown in the next section how to apply this approach to the analysis of the actual overall domain including its heterogeneous part.

3.1 *Elastodynamic problem in the structure*

The Navier equation in the structural domain for the Green function $\mathbf{u}^0{}_g(\mathbf{x},\omega;\boldsymbol{\xi},\mathbf{e})$ computed in the structure when the unit load is exerted in the soil writes:

$$\mathbf{Div}\boldsymbol{\sigma}_b(\mathbf{u}_g^0) = -\omega^2 \rho_b \mathbf{u}_g^0 \qquad \text{on } \Omega_b \tag{5}$$

The solution of this local problem is expanded in the form:

$$\mathbf{u}_g^0(\mathbf{x},\omega) = \sum_{p=1}^{P} c_p(\omega)\mathbf{L}^*_p(\mathbf{x},\omega) + \sum_{A=1}^{N} \alpha_A(\omega)\boldsymbol{\phi}_A(\mathbf{x},\omega) \qquad \forall \mathbf{x} \in \Omega_b, \forall \omega \in 3 \tag{6}$$

$\{\mathbf{L}^*_p,\ p = 1, \ldots P\}$ are the static modes of the structure, their traces on the interface Γ being the interface basis noted $\{\mathbf{L}_p,\ p = 1, \ldots P\}$; $\{\boldsymbol{\phi}_A,\ A = 1, \ldots N\}$ are the fixed basis eigen-modes of the structure with eigen-values $\{\omega_A\}$ and critical dampings $\{\xi_A\}$.

3.2 *Elastodynamic problem in the reference soil domain*

The Navier equation in the reference soil domain for the Green function $\mathbf{u}^0{}_g(\mathbf{x},\omega;\boldsymbol{\xi},\mathbf{e})$ computed in the soil when the unit load is exerted in it writes:

$$\begin{aligned} &\mathbf{Div}\boldsymbol{\sigma}_{s0}(\mathbf{u}_g^0) + \mathbf{q}(\boldsymbol{\xi}) = -\omega^2 \rho_{s0} \mathbf{u}_g^0 \qquad \text{on } \Omega_{s0} \\ &\mathbf{q}(\boldsymbol{\xi}) = \delta(\mathbf{x} - \boldsymbol{\xi})\mathbf{e} \end{aligned} \tag{7}$$

Due to the presence of the structure, $\mathbf{u}^0{}_g$ is decomposed into an elementary incident field $\mathbf{u}^0{}_q(\mathbf{x},\omega)$, a locally scattered field $\mathbf{u}^0{}_{qd}(\mathbf{x},\omega)$ and a diffracted field defined below, as:

$$\mathbf{u}_g^0(\mathbf{x},\omega) = \mathbf{u}_q^0(\mathbf{x},\omega) + \mathbf{u}_{qd}^0(\mathbf{x},\omega) + \mathbf{u}_{d*}^0(\mathbf{x},\omega) \qquad \forall \mathbf{x} \in \Omega_{s0}, \forall \omega \in 3 \tag{8}$$

where $\mathbf{u}^0{}_{qd}(\mathbf{x},\omega)$ satisfies the homogeneous Navier equation in Ω_{s0}, the Sommerfeld's radiation conditions on $\Gamma_{s\infty}$ and the boundary conditions:

$$\begin{aligned} &\mathbf{u}_{qd}^0 = -\mathbf{u}_q^0 \qquad \text{on } \Gamma \\ &\mathbf{t}_\mathbf{n}(\mathbf{u}_{qd}^0) = \mathbf{0} \qquad \text{on } \Gamma_{as} \end{aligned} \tag{9}$$

The diffracted field satisfies the homogeneous Navier equation in Ω_{s0}, the Sommerfeld's radiation conditions on $\Gamma_{s\infty}$ and the boundary conditions:

$$\begin{aligned} &\mathbf{u}_{d*}^0 = \sum_{p=1}^{P} c_p \mathbf{L}_p \qquad \text{on } \Gamma \\ &\mathbf{t}(\mathbf{u}_{d*}^0) = \mathbf{0} \qquad \text{on } \Gamma_{as} \end{aligned} \tag{10}$$

The elementary incident field satisfies the heterogeneous Navier equation in Ω_{s0}, the Sommerfeld's radiation conditions on $\Gamma_{s\infty}$ and the traction-free boundary condition:

$$\begin{aligned} &\mathbf{t}(\mathbf{u}_q^0) = \mathbf{0} \qquad \text{on } \Gamma_{as} \\ &\mathbf{Div}\boldsymbol{\sigma}_{s0}(\mathbf{u}_q^0) + \mathbf{q} = -\omega^2 \rho_{s0}(\mathbf{u}_q^0) \qquad \text{in } \Omega_{s0} \end{aligned} \tag{11}$$

We now introduce the elementary diffracted displacement fields $\{\mathbf{u}^0_{dp}, p = 1, \ldots P\}$ which satisfy the homogeneous Navier equation in the reference soil domain, the Sommerfeld's radiation conditions on $\Gamma_{s\infty}$ and the boundary conditions:

$$\begin{cases} \mathbf{u}_{dp}^0 = \mathbf{L}_p, \; p = 1, \ldots P \qquad \text{on } \Gamma \\ \mathbf{t}(\mathbf{u}_{dp}^0) = \mathbf{0}, \; p = 1, \ldots P \qquad \text{on } \Gamma_{as} \\ \mathbf{Div}\boldsymbol{\sigma}_{s0}(\mathbf{u}_{dp}^0) = -\omega^2 \rho_{s0} \mathbf{u}_{dp}^0, \; p = 1, \ldots P \qquad \text{in } \Omega_{s0} \end{cases} \tag{12}$$

These fields are computed by boundary element equations for a geometrical discretization of the soil-structure interface which are described in the following section. Obviously the total displacement field in the soil domain may now be expanded on this basis with the unknown participation factors $\{c_p, p = 1, \ldots P\}$ as:

$$\mathbf{u}_g^0 = \mathbf{u}_q^0 + \mathbf{u}_{qd}^0 + \sum_{p=1}^{P} c_p \mathbf{u}_{dp}^0 \tag{13}$$

3.3 *Boundary integral equations for the reference soil medium*

For any field $\mathbf{u}^0_d$ ($\mathbf{u}^0_{dp}$ or $\mathbf{u}^0_{qd}$) satisfying the homogeneous Navier equation in the reference soil domain and the Sommerfeld's radiation conditions, one has the following direct boundary integral equation for Γ assumed regular:

$$\begin{aligned} &\int_\Gamma \mathbf{U}^G(\mathbf{x}, \mathbf{x}_\Gamma)\mathbf{t}^0(\mathbf{u}_d^0)(\mathbf{x}_\Gamma) - \mathbf{T}^G(\mathbf{x}, \mathbf{x}_\Gamma)\mathbf{u}_d^0(\mathbf{x}_\Gamma)\, dS(\mathbf{x}_\Gamma) = \boldsymbol{\kappa}(\mathbf{x})\mathbf{u}_d^0(\mathbf{x}) \qquad p = 0, \ldots P \\ &\boldsymbol{\kappa}(\mathbf{x}) = \frac{1}{2}\mathbf{I} \qquad \mathbf{x} \in \Gamma / \Gamma_{as} \\ &\boldsymbol{\kappa}(\mathbf{x}) = \mathbf{I} \qquad \mathbf{x} \in (\Gamma \cap \Gamma_{as}) \cup \Omega_{s0} \\ &\boldsymbol{\kappa}(\mathbf{x}) = \mathbf{0} \qquad \mathbf{x} \notin \overline{\Omega_{s0}} \end{aligned} \tag{14}$$

where $\mathbf{U}^G$ and $\mathbf{T}^G$ are the classical first and second Green tensors for the reference soil medium *alone* and $\boldsymbol{\kappa}$ is the so called free term. For $\mathbf{u}^0_q$, we simply have:

$$\mathbf{u}_q^0(\mathbf{x}) = \mathbf{U}^G(\mathbf{x}, \boldsymbol{\xi})\mathbf{e} \qquad \forall \mathbf{x} \in \overline{\Omega_{s0}} \tag{15}$$

In order to solve these problems numerically, given the boundary conditions of (9) and (12), the interface Γ is meshed into N_Γ elements and the fields $\mathbf{L}_p$, $\mathbf{u}^0_{qd}$, and $\mathbf{t}^0(\mathbf{u}^0_{dp})$, $\mathbf{t}^0(\mathbf{u}^0_{qd})$ are interpolated on constant shape functions over Γ. Given the interpolation coefficients of $\mathbf{L}_p$, $p = 1, \ldots P$, Equation (14) gives a way to calculate the interpolation coefficients of $\mathbf{t}^0(\mathbf{u}^0_{dp})$ and the values of the elementary diffracted fields $\mathbf{u}^0_{dp}$ at any points in the soil reference soil medium. It is solved similarly for $\mathbf{u}^0_{qd}$.

Several numerical difficulties arise from these equations. First, the integral operator with first Green's tensor kernel is weakly singular in $1/|\mathbf{x} - \boldsymbol{\xi}|$ so that a particular integration scheme is required: it consists in computing these integrals in cylindrical coordinates (see Bonnet 1995) for the surface boundary elements decomposed in sub-elements. Secondly, the integral operator with second Green's tensor kernel is strongly singular in $1/|\mathbf{x} - \boldsymbol{\xi}|^2$ and a regularization technique is applied in this case (Aubry & Clouteau 1991): the static and dynamic Green tensors having the same singularity as $|\mathbf{x} - \xi| \to 0$, a weakly singular boundary integral equation is obtained by subtracting the static equation from the dynamic one. Finally, the introduction of some damping in the soil makes up for the problem of the irregular frequencies inherent in boundary integral formulations.

3.4 *Coupled solution in the reference medium*

The participation coefficients α_A and c_p which now entirely define the Green function of the overall reference domain, are derived from the equilibrium of the structure written using the principle of virtual work:

$$\left\langle \mathbf{t}_b(\mathbf{u}_g^0) + \mathbf{t}_s^0(\mathbf{u}_g^0), \mathbf{v} \right\rangle_\Gamma = \left(\boldsymbol{\sigma}_b(\mathbf{u}_g^0), \boldsymbol{\varepsilon}(\mathbf{v}) \right)_{\Omega_b} - \omega^2 \left(\rho_b \mathbf{u}_g^0, \mathbf{v} \right)_{\Omega_b} \tag{16}$$

for any structural eigen-mode or static mode $\mathbf{v}$. This leads to the algebraic system:

$$\begin{bmatrix} \mathbf{K}_b^{\phi\phi} & -\omega^2 \mathbf{M}_b^{\phi L} \\ -\omega^2 \mathbf{M}_b^{L\phi} & \mathbf{K}_b^{LL} + \mathbf{K}_s^0 \end{bmatrix} \begin{pmatrix} \boldsymbol{\alpha} \\ \mathbf{c}_\Gamma \end{pmatrix} = \begin{pmatrix} \mathbf{0} \\ \mathbf{F}_q^0 \end{pmatrix} \tag{17}$$

where:

$$[\mathbf{K}_b^{\phi\phi}]_{AB} = (\omega_A^2 - \omega^2 + 2i\xi_A \omega_A \omega)\delta_{AB} \qquad A, B = 1,\ldots N \tag{17a}$$

$$[\mathbf{K}_b^{LL}]_{pp'} = -\omega^2 \left(\rho_b \mathbf{L}^*_p, \mathbf{L}^*_{p'} \right)_{\Omega_b} + \left(\boldsymbol{\sigma}(\mathbf{L}^*_p), \boldsymbol{\varepsilon}(\mathbf{L}^*_{p'}) \right)_{\Omega_b} \qquad p, p' = 1,\ldots P \tag{17b}$$

$$[\mathbf{M}_b^{\phi L}]_{Ap} = \left(\rho_b \boldsymbol{\phi}_A, \mathbf{L}^*_p \right)_{\Omega_b} = [\mathbf{M}_b^{L\phi}]_{pA} \qquad A = 1,..N;\ p = 1,\ldots P \tag{17c}$$

$$[\mathbf{K}_s^0]_{pp'} = \left\langle \mathbf{t}^0(\mathbf{u}_{dp}^0), \mathbf{L}_{p'} \right\rangle_\Gamma \qquad p, p' = 1,\ldots P \tag{17d}$$

$$[\mathbf{F}_q^0]_p = -\left\langle \mathbf{t}_s^0(\mathbf{u}_q^0 + \mathbf{u}_{qd}^0), \mathbf{L}_p \right\rangle_\Gamma \tag{17e}$$

and $\boldsymbol{\alpha}$ and $\mathbf{c}_\Gamma$ are the vectors of the participation factors; $\mathbf{K}^0_s$ denotes the frequency dependent impedance matrix of the foundation resting on the reference soil.
Remark: The field diffracted by the foundation movement when it is impinged by a seismic incident field $\mathbf{u}^0_i$ propagating in the reference soil medium is computed exactly in the same way substituting $\mathbf{u}^0_q$ by $\mathbf{u}^0_i$.

4 COUPLING WITH THE HETEROGENEOUS SOIL PART

4.1 *Displacement fields*

The Navier equation in the heterogeneous soil part Ω_{s1} is written:

$$\mathbf{Div}\boldsymbol{\sigma}_{s1}(\mathbf{u}_s) = -\omega^2 \rho_{s1} \mathbf{u}_s + \boldsymbol{\psi} \qquad \text{on } \Omega_{s1} \tag{18}$$

so that, with the definition given above to the force field $\boldsymbol{\psi}$, the total displacement field $\mathbf{u}_s$ in the actual soil under a seismic incident field $\mathbf{u}^0_i$ and including the effects of interaction with the structure, is given by:

$$\mathbf{u}_s(\mathbf{x}_s, \omega) = \mathbf{u}_i^0(\mathbf{x}_s, \omega) + \mathbf{u}_{id}^0(\mathbf{x}_s, \omega) + \sum_{k=1}^{3} \int_{\Omega_{s1}} \mathbf{u}_g^0(\mathbf{x}_s, \omega; \mathbf{x}, \mathbf{e}_k)(\boldsymbol{\psi}(\mathbf{x}).\mathbf{e}_k)dV(\mathbf{x}) \tag{19}$$

In this equation, $\mathbf{u}^0_{id}$ is the total field diffracted by the foundation movement in the reference soil domain when the seismic incident field considered impinges it.

The force field $\boldsymbol{\psi}$ is now expanded as a linear combination of $3N_1$ elementary force fields as:

$$\boldsymbol{\psi} = \sum_{Ek=1,1}^{3N_1} \beta_{Ek}(\omega)\boldsymbol{\psi}_{Ek}(\mathbf{x},\omega) \tag{20}$$

when the perturbed soil part Ω_{s1} is discretized by N_1 finite elements. The new participation factors introduced in (20) are determined using the principle of virtual works applied to Equation (18) for any displacement test field $\mathbf{v}$ chosen among all admissible displacement fields. Injecting (19) into (18), the following system is obtained:

$$\left[\mathbf{K}^0_\Omega - \mathbf{K}_1\right](\boldsymbol{\beta}) = (\mathbf{F}_1) \tag{21}$$

with:

$$\left[\mathbf{K}^0_\Omega\right]_{EkFl} = -\left(\boldsymbol{\psi}_{Ek}, \Delta\mathbf{u}^0_{Fl}\right)_{\Omega_{s1}} \tag{21a}$$

$$\left[\mathbf{K}_1\right]_{EkFl} = -\omega^2\left(\rho_1\Delta\mathbf{u}^0_{Ek}, \Delta\mathbf{u}^0_{Fl}\right)_{\Omega_{s1}} + \left(\boldsymbol{\sigma}_1(\Delta\mathbf{u}^0_{Ek}), \boldsymbol{\varepsilon}(\Delta\mathbf{u}^0_{Fl})\right)_{\Omega_{s1}} \tag{21b}$$

$$\left[\mathbf{F}_1\right]_{Ek} = -\omega^2\left(\rho_1(\mathbf{u}^0_i + \mathbf{u}^0_{id}), \Delta\mathbf{u}^0_{Ek}\right)_{\Omega_{s1}} + \left(\boldsymbol{\sigma}_1(\mathbf{u}^0_i + \mathbf{u}^0_{id}), \boldsymbol{\varepsilon}(\Delta\mathbf{u}^0_{Ek})\right)_{\Omega_{s1}} \tag{21c}$$

where $\boldsymbol{\beta}$ is the vector of the participation factors β_{Ek}, and $\Delta\mathbf{u}^0$ is defined by:

$$\Delta\mathbf{u}^0_{Ek}(\mathbf{x}_s,\omega) = \int_E \mathbf{u}^0_g(\mathbf{x}_s,\omega;\mathbf{x}_E,\mathbf{e}_k)(\boldsymbol{\psi}_{Ek}(\mathbf{x}_E).\mathbf{e}_k)dV(\mathbf{x}_E) \qquad E \in \{1,\ldots N_1\}, k \in \{1,2,3\} \tag{21d}$$

In the above, $\mathbf{K}^0_\Omega$ is the impedance operator of the overall reference domain seen from the heterogeneity Ω_{s1}. Once the participation factors have been computed by Equation (21), the total displacement field in the actual soil domain is obtained by Equation (19). At last, the time domain results are derived applying a numerical inverse Fourier Transform. The total displacement field in the structure at a point $\mathbf{x}_b$ is given similarly:

$$\mathbf{u}_b(\mathbf{x}_b) = \mathbf{u}^0_b(\mathbf{x}_b) + \sum_{k=1}^{3}\int_{\Omega_{s1}} \mathbf{u}^0_g(\mathbf{x}_b,\omega;\mathbf{x},\mathbf{e}_k)(\boldsymbol{\psi}(\mathbf{x}).\mathbf{e}_k)dV(\mathbf{x}) \tag{22}$$

where $\mathbf{u}^0_b$ is the displacement field in the structure in the absence of the heterogeneous part.

4.2 *Impedance functions*

The computation of impedance functions for the foundation resting on the actual soil (including the heterogeneous part) can be achieved by a similar approach as it is shown now. They are given by Equation (17d) for the foundation resting on the reference soil. Similarly, for the actual soil domain they are:

$$[\mathbf{K}_s]_{pp'} = \left\langle \mathbf{t}(\mathbf{u}_{dp}), \mathbf{L}_{p'}\right\rangle_\Gamma \qquad p, p' = 1,\ldots P \tag{23}$$

where the elementary diffracted displacement fields $\{\mathbf{u}_{dp}, p = 1, \ldots P\}$ satisfy the homogeneous Navier equation in the actual soil domain Ω_s, the Sommerfeld's radiation conditions on $\Gamma_{s\infty}$ and the boundary conditions:

$$\begin{cases} \mathbf{u}_{dp} = \mathbf{L}_p,\ p = 1,\ldots P & \text{on } \Gamma \\ \mathbf{t}(\mathbf{u}_{dp}) = \mathbf{0},\ p = 1,\ldots P & \text{on } \Gamma_{as} \\ \mathbf{Div}\boldsymbol{\sigma}_s(\mathbf{u}_{dp}) = -\omega^2\rho_s\mathbf{u}_{dp},\ p = 1,\ldots P & \text{in } \Omega_s \end{cases} \tag{24}$$

Multiplying the homogeneous Navier equation in (12) written for $\mathbf{u}^0_{dp'}$ by $\mathbf{u}_{dp}$ and the homogeneous Navier equation in (24) written for $\mathbf{u}_{dp}$ by $\mathbf{u}^0_{dp'}$, integrating them by parts over Ω_s and subtracting them, leads to the following expression for the impedance functions on the actual soil:

$$[\mathbf{K}_s]_{pp'} = [\mathbf{K}^0_s]_{pp'} + \left(\boldsymbol{\sigma}_1(\mathbf{u}_{dp}), \boldsymbol{\varepsilon}(\mathbf{u}^0_{dp'})\right)_{\Omega_{s1}} - \omega^2 \left(\rho_1 \mathbf{u}_{dp}, \mathbf{u}^0_{dp'}\right)_{\Omega_{s1}} \qquad p, p' = 1,...P \tag{25}$$

The elementary diffracted displacement fields $\{\mathbf{u}_{dp},\ p = 1, \dots P\}$ in the actual soil domain are written in accordance to Equations (19) and (21d):

$$\mathbf{u}_{dp}(\mathbf{x}_s, \omega) = \mathbf{u}^0_{dp}(\mathbf{x}_s, \omega) + \sum_{Ek=1,1}^{3N_1} \beta^{(p)}_{Ek} \Delta \mathbf{u}^0_{Ek} \tag{26}$$

where the vector $\boldsymbol{\beta}^{(p)}$ of the corresponding participation factors $\beta^{(p)}_{Ek}$ is obtained as the solution of a problem similar to Equation (21) for the Green function $\mathbf{u}^0_g(\mathbf{x},\omega;\boldsymbol{\xi},\mathbf{e})$ computed in the absence of the superstructure. If we note $\boldsymbol{\psi}_p$ the force field then obtained:

$$\boldsymbol{\psi}_p = \sum_{Ek=1,1}^{3N_1} \beta^{(p)}_{Ek} \boldsymbol{\psi}_{Ek} \tag{27}$$

the impedance functions are given by:

$$[\mathbf{K}_s]_{pp'} = [\mathbf{K}^0_s]_{pp'} + \left(\boldsymbol{\psi}_p, \mathbf{u}^0_{dp'}\right)_{\Omega_{s1}} \qquad p, p' = 1,...P \tag{28}$$

5 RANDOM HETEROGENEITY

We know focus on the case of a random perturbed soil domain characterized by the random field $\boldsymbol{\gamma} = (\rho_{s1},\lambda_{s1},\mu_{s1})$. The method proposed to address this problem is to introduce a stochastic linearization of this vector on the basis of the eigen-modes of its covariance operator. This defines the Karhunen-Loeve expansion of the random field $\boldsymbol{\gamma}$ (see Soize 1993 for details) and Spanos & Ghanem (1991) have called its application to the finite elements technique the spectral finite elements method. We first introduce the basic hypothesis required for the random field, and then implement the method in the elastodynamic problem described above.

5.1 *Hypothesis for the random perturbed soil modulii*

$(\rho_{s1},\lambda_{s1},\mu_{s1})$ for the heterogeneous soil part are modeled by a second-ordered non-homogeneous centered stochastic field $\boldsymbol{\gamma}(\mathbf{x})$ indexed on Ω_{s1} with values, for generality purposes, in $\forall^3$ to account for damping. We suppose that all realizations of $\boldsymbol{\gamma}(\mathbf{x})$ are square integrable over Ω_{s1}. The covariance function of $\boldsymbol{\gamma}$ is denoted $\mathbf{C}_\gamma$ and it is defined on $\Omega_{s1}\times\Omega_{s1}$ with values in $\mathrm{Mat}_\forall(3,3)$ by:

$$\mathbf{C}_\gamma(\mathbf{x},\mathbf{x}') = E\left\{\boldsymbol{\gamma}(\mathbf{x}) \otimes \overline{\boldsymbol{\gamma}(\mathbf{x}')}\right\} \tag{29}$$

where E denotes mathematical expectation. It is associated with the covariance operator $\boldsymbol{\mathcal{C}}_\gamma$ defined for every $\mathbf{v}_1$ and $\mathbf{v}_2$ in $\forall^3$ by:

$$\left(\boldsymbol{\mathcal{C}}_\gamma \mathbf{v}_1, \mathbf{v}_2\right) = \int_{\Omega_{s1}} \int_{\Omega_{s1}} \mathbf{C}_\gamma(\mathbf{x},\mathbf{x}')\mathbf{v}_1(\mathbf{x}').\overline{\mathbf{v}_2(\mathbf{x})}\, dV(\mathbf{x}')\, dV(\mathbf{x}) \tag{30}$$

Under the above assumption for the stochastic field $\boldsymbol{\gamma}$, $\boldsymbol{\mathcal{C}}_\gamma$ has a discrete set of eigen-vectors $\boldsymbol{\gamma}_k$ and eigen-values χ_k such that the covariance function can be expanded in the form:

$$\mathbf{C}_\gamma(\mathbf{x},\mathbf{x}') = \sum_{k=1}^{\infty} \chi_k^2 \boldsymbol{\gamma}_k(\mathbf{x}) \otimes \overline{\boldsymbol{\gamma}_k(\mathbf{x}')} \tag{31}$$

with:

$$(\boldsymbol{\gamma}_k, \boldsymbol{\gamma}_l) = \delta_{kl} \tag{32}$$

Then the stochastic field $\boldsymbol{\gamma}$ is given by:

$$\boldsymbol{\gamma}(\mathbf{x}) = \sum_{k=1}^{\infty} \theta_k \chi_k \boldsymbol{\gamma}_k(\mathbf{x}) \tag{33}$$

where the θ_k are uncorrelated and centered random variables in $\forall$ defined by the projection of the stochastic field on the eigen-modes of its covariance operator. They have the orthogonality property:

$$\begin{aligned} &E\{\theta_k\} = 0 \\ &E\{\theta_k \overline{\theta_l}\} = \delta_{kl} \end{aligned} \tag{34}$$

If the field $\boldsymbol{\gamma}(\mathbf{x})$ is Gaussian, these random variables are also Gaussian, as linear transformations of a Gaussian process. They are also mutually independent.

Practically, the eigen-vectors of the covariance operator are computed up to an order r. The choice of the expansion order is directly imposed by the relative precision required in the analysis. Let P_γ be the mean square value of the random field $\boldsymbol{\gamma}$, it is given by:

$$P_\gamma = E\left\{\|\boldsymbol{\gamma}\|^2_{\Omega_{s1}}\right\} = \sum_{k=1}^{\infty} \chi_k^2 = \mathrm{tr}\boldsymbol{C}_\gamma \tag{35}$$

If we denote by $\boldsymbol{\gamma}_r$ the random field obtained through the expansion of Equation (33) up to the order r, the relative tolerance of this approximation ε is chosen such that:

$$E\left\{\|\boldsymbol{\gamma} - \boldsymbol{\gamma}_r\|^2_{\Omega_{s1}}\right\} = \sum_{k=r+1}^{\infty} \chi_k^2 \le \varepsilon P_\gamma , \tag{36}$$

which gives the value of r for this tolerance using the following inequation:

$$1 - \frac{\sum_{k=1}^{r} \chi_k^2}{P_\gamma} \le \varepsilon \tag{37}$$

For a given choice of the relative tolerance, the expansion order r is not too large as long as the spatial correlation lengths of the field $\boldsymbol{\gamma}$ are not too small, that is the covariance function needs not to tend to zero too fast for the method to be efficient. However, it is not possible in the general cases to obtain a more precise information on the decreasing speed of the eigen-values of the covariance operator, or a criterion to fix r versus the shape of the covariance function.

5.2 *Application to the discretized dynamic stiffness operator of the heterogeneous part of the soil*

The dynamic stiffness operator $\mathbf{K}_1$ of the heterogeneous part of the soil Ω_{s1} is defined by:

$$(\mathbf{K}_1\mathbf{u}, \mathbf{v}) = -\omega^2 (\rho_1 \mathbf{u}, \mathbf{v})_{\Omega_{s1}} + (\boldsymbol{\sigma}_1(\mathbf{u}), \boldsymbol{\varepsilon}(\mathbf{v}))_{\Omega_{s1}} \tag{38}$$

and has been discretized on a finite element mesh with N_1 elements and M nodes. The covariance operator is also discretized on this mesh, and the random field $\boldsymbol{\gamma}$ is evaluated by the expansion (33) up to an order $r \le 3M$. The dynamic stiffness operator is then expanded:

$$\mathbf{K}_1(\omega) = \sum_{k=1}^{r} \theta_k \mathbf{K}_k(\omega) \tag{39}$$

with obvious notations for the deterministic operators $\mathbf{K}_k$ computed from the eigen-vectors of the covariance operator of the soil random modulii. A similar expansion is written for $\mathbf{F}_1$. If P_γ^M is the mean square value of the random field $\boldsymbol{\gamma}$ for the finite element mesh considered:

$$P_\gamma^M = \sum_{k=1}^{3M} \chi_k^2 \ , \tag{40}$$

it is computed as the trace of the discretized covariance operator and the relative tolerance chosen in the analysis imposes the truncation order r through:

$$1 - \frac{\sum_{k=1}^{r} \chi_k^2}{P_\gamma^M} \leq \varepsilon \tag{41}$$

The expansion (39) of the dynamic stiffness operator is know injected in Equation (21) whose solution is obtained by a Monte-Carlo simulation technique in order to derive the reference solution for the problem considered. For small random perturbations of the soil parameters, that is if the random field $\boldsymbol{\gamma}$ is small compared to the parameters of the reference soil medium, a Neumann expansion can be introduced in combination with the Karhunen-Loeve expansion.

5.3 *Solution by Neumann expansion*

We now multiply both sides of Equation (21) by $(\mathbf{K}^0_\Omega)^{-1}$ and note:

$$\begin{aligned} \tilde{\mathbf{K}}_1 &= (\mathbf{K}_\Omega^0)^{-1}\mathbf{K}_1 \\ \tilde{\mathbf{F}}_1 &= (\mathbf{K}_\Omega^0)^{-1}\mathbf{F}_1 \end{aligned} \tag{42}$$

It is assumed that the random perturbed modulii remain small compared to the modulii of the reference soil, which writes:

$$\operatorname*{Sup}_{\|\mathbf{u}\|\leq 1} \left\| \tilde{\mathbf{K}}_1 \mathbf{u} \right\| < 1 \tag{43}$$

Then by a Neumann expansion we have:

$$(\mathbf{I} - \tilde{\mathbf{K}}_1)^{-1} = \sum_{i=1}^{\infty} (\tilde{\mathbf{K}}_1)^i \tag{44}$$

This expansion is used in combination with the Karhunen-Loeve expansions introduced above to give the vector of the participation factors as:

$$\boldsymbol{\beta} = \sum_{i=1}^{\infty} \left(\sum_{k=1}^{r} \theta_k \tilde{\mathbf{K}}_k \right)^i \sum_{k=1}^{r} \theta_k \tilde{\mathbf{F}}_k \tag{45}$$

The second order statistics of $\boldsymbol{\beta}$, namely its mean and covariance, can be computed from the above expression written up to the order $i = R$ provided the cross-correlation functions of the random variables θ_k are known. This is hardly the case, except if they are Gaussian and real; then the following properties hold $\forall n \in \angle^*$:

$$
\begin{aligned}
&E\left\{\prod_{i=1}^{2n+1}\theta_{k_i}\right\}=0 \\
&E\left\{\prod_{i=1}^{2n}\theta_{k_i}\right\}=\sum_{j=2}^{2n}\left[E\left\{\theta_{k_1}\theta_{k_j}\right\}E\left\{\prod_{i\neq j}\theta_{k_i}\right\}\right]
\end{aligned}
\tag{46}
$$

For R = 2, algebraic expressions for the mean and cross-correlation function of $\boldsymbol{\beta}$ can be worked out as:

$$
\begin{aligned}
&E\{\boldsymbol{\beta}\}=\sum_{k=1}^{r}\tilde{\mathbf{K}}_k\tilde{\mathbf{F}}_k \\
&E\left\{\boldsymbol{\beta}\otimes\bar{\boldsymbol{\beta}}\right\}=\sum_{k=1}^{r}\tilde{\mathbf{F}}_k\overline{\tilde{\mathbf{F}}_k^T} \\
&\qquad+\sum_{k=1}^{r}\sum_{l=1}^{r}\sum_{m=1}^{r}\sum_{n=1}^{r}E\{\theta_k\theta_l\theta_m\theta_n\}\left[\tilde{\mathbf{K}}_k\tilde{\mathbf{F}}_l\overline{\tilde{\mathbf{F}}_m^T\tilde{\mathbf{K}}_n}+\tilde{\mathbf{F}}_k\overline{\tilde{\mathbf{F}}_l^T\tilde{\mathbf{K}}_m\tilde{\mathbf{K}}_n}+\tilde{\mathbf{K}}_k\tilde{\mathbf{K}}_l\tilde{\mathbf{F}}_m\overline{\tilde{\mathbf{F}}_n^T}\right] \\
&\text{with } E\{\theta_k\theta_l\theta_m\theta_n\}=\delta_{kl}.\delta_{mn}+\delta_{km}.\delta_{ln}+\delta_{kn}.\delta_{lm}
\end{aligned}
\tag{47}
$$

The above equations show that for higher orders the second order statistics derived from such an analysis technique are hardly tractable from algebraic developments, although important simplifications can be obtained thanks to Equations (46) for a Gaussian real-valued random field $\boldsymbol{\gamma}$. In general cases and for orders higher than 2, numerical simulations are needed if a good precision has to be reached.

The procedure described above has several advantages: first, Karhunen-Loeve reduces the "randomness" of the field $\boldsymbol{\gamma}$ to a set of stochastic numbers so that in Monte-Carlo procedures draws on random variables rather than simulations of fields have to be performed. Secondly, in this approach the computation of the Green function for the overall reference domain and the inversion of its impedance function seen from the heterogeneity have only to be done once.

6 CONCLUSIONS

An original method to account for the effects of spatial variability of the soil modulii on seismic soil-structure interaction based on the sub-structuring technique and the boundary element method has been presented. It requires the preliminary computation of the Green function of the overall unperturbed reference domain constituted by the structure and the soil without its bounded perturbed part, as well as its impedance functions when it is seen from the domain occupied by the perturbation. Then the final solution has been obtained writing the equilibrium of the perturbed soil part. Expressions for the displacement fields in the soil and in the structure have been derived. Other relations of great interest in engineering applications are given for the impedance functions of the foundation resting on the actual soil medium with its perturbation. These results are intended to be applied to soil modulii updating from forced vibrations experiments. Further developments are however still needed and will be reported subsequently.

The formulation allows introducing deterministic or random perturbed soil modulii. In the latter case, the Karhunen-Loeve expansion of the perturbed soil parameters has been used. It consists in projecting them on the eigen-vectors basis of their covariance operator. The method is very efficient if the spatial correlation lengths of the random parameters are not too small, since it reduces their randomness dimension to a relatively small set of random numbers instead of random fields. For small perturbations with respect to the reference soil medium, a Neumann expansion has been implemented and, up to the second order of the expansion, algebraic expressions for the second order statistics of the problem solution have been given.

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